

Alex Supertramp Osbourne

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Senior Backend Engineer | Cloud Infrastructure | Distributed Systems | Platform Engineering

PROFESSIONAL SUMMARY

Senior Backend Engineer with 8+ years building and scaling distributed systems for data-intensive products. Currently leading platform reliability efforts at Skyline Analytics, designing event-driven services that process millions of records daily while keeping the team's deploy frequency high and incident rate low.

Comfortable across the stack from API design to infrastructure: Go and Python services, Kafka-based pipelines, PostgreSQL and Redis, deployed on Kubernetes across multiple cloud providers. Strong advocate for observability-first design, incremental migrations, and pairing with product teams to turn ambiguous requirements into shippable systems. Previously built payment processing services in a regulated fintech environment (PCI DSS compliance, idempotent transaction handling, audit logging) and led the migration of a monolithic e-commerce platform to a service-oriented architecture, cutting average page load time by 40% and enabling independent team deploys.

Enjoys mentoring engineers, running architecture reviews, and writing internal tooling that removes toil from everyday development work.

PROFESSIONAL EXPERIENCE

Senior Backend Engineer

Skyline Analytics | Mar 2022 - Present

Own the ingestion and processing pipeline for a real-time analytics platform, handling data from thousands of customer integrations. Lead reliability and scalability initiatives for the core event-processing services.

- Redesigned the event ingestion pipeline around Kafka and Go consumers, increasing sustained throughput from 8k to 45k events/sec while cutting p99 processing latency by 60%
- Led the migration from a single Postgres instance to a sharded, read-replica architecture, eliminating recurring write contention incidents during peak load
- Introduced structured tracing and metrics (OpenTelemetry, Prometheus, Grafana) across 12 services, cutting mean time to detect production issues from ~40 minutes to under 5
- Designed a backpressure-aware retry and dead-letter-queue system for downstream consumers, removing the need for manual replay during partner outages
- Mentored 3 mid-level engineers through their first on-call rotations and led biweekly architecture review sessions for the platform team
- Migrated CI/CD from Jenkins to GitHub Actions with reusable workflows, reducing average deploy time from 25 minutes to 7

Technologies used: Kafka, Go, distributed systems, PostgreSQL, database sharding, observability, OpenTelemetry, Prometheus, Grafana, reliability, CI/CD, GitHub Actions, technical leadership

Backend Engineer

Northbridge Payments | Jun 2019 - Feb 2022

Built and maintained core transaction processing services for a payments platform operating under PCI DSS compliance requirements, focusing on correctness, auditability, and idempotency guarantees.

- Designed and implemented an idempotent transaction API used by all downstream payment flows, eliminating a class of double-charge incidents reported by merchants
- Built an append-only audit logging system for all transaction state changes, satisfying PCI DSS traceability requirements ahead of the annual compliance audit
- Led the rollout of contract testing (Pact) between the payments service and 6 consuming teams, catching breaking API changes in CI before they reached staging
- Optimized the settlement batch job from a 3-hour nightly run to under 20 minutes by parallelizing reconciliation across merchant shards
- Served as on-call lead for the payments team, authoring runbooks that cut average incident resolution time by half

Technologies used: Java, Spring Boot, payments, PCI DSS, compliance, contract testing, Pact, performance optimization, incident management, reliability

Software Engineer

Greenfield Retail | Aug 2017 - May 2019

Contributed to the migration of a monolithic e-commerce platform into independently deployable services, working across backend APIs and customer-facing storefront features.

- Extracted the product catalog and search functionality from the monolith into a standalone Node.js service backed by Elasticsearch, cutting search response times by 70%
- Built React-based storefront features including faceted search, cart persistence, and a checkout flow redesign that reduced cart abandonment by 12%
- Implemented a feature-flagging system enabling gradual rollouts, reducing the blast radius of the 6 major releases shipped during the migration
- Cut average page load time by 40% across the storefront through API response caching and image delivery optimization

Technologies used: Node.js, Elasticsearch, microservices, React, e-commerce, feature flags, release management, performance, caching, CDN

Junior Developer

Cedar Labs | Jul 2015 - Jul 2017

Built and maintained internal tools and small customer-facing features for a workflow automation product, working closely with senior engineers to learn production engineering practices.

- Built an internal admin dashboard for support staff to inspect and replay failed customer workflows, reducing average ticket resolution time by 30%
- Wrote the team's first suite of automated integration tests for the workflow engine, catching regressions before they reached production
- Maintained and extended a PHP/MySQL backend supporting the core workflow automation product

Technologies used: PHP, MySQL, JavaScript, internal tools, testing, integration tests, quality assurance

PROJECTS

Open-Source Rate Limiter (Go)

2023 - Present

A distributed token-bucket rate limiter used by several small SaaS backends, published as an open-source Go module.

- Implemented a Redis-backed sliding-window algorithm supporting 50k+ requests/sec with sub-millisecond overhead
- Published as an open-source module with 200+ GitHub stars and adopted by 3 external teams
- Wrote benchmark suites comparing token-bucket and sliding-window strategies under bursty load

Technologies used: Go, Redis, distributed systems, open source, concurrency

Personal Finance Dashboard

2021 - 2022

A self-hosted dashboard aggregating bank and investment data via the Plaid API, built to learn React and time-series visualization.

- Built a Plaid integration syncing transactions across 4 account types nightly
- Designed interactive D3 charts for spending trends and net-worth tracking

Technologies used: React, TypeScript, Plaid API, data visualization

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate - Amazon Web Services (AWS)
- AWS Certified Developer - Associate - Amazon Web Services (AWS)
- KCNA: Kubernetes and Cloud Native Associate - The Linux Foundation / CNCF
- Certified Kubernetes Administrator (CKA) - The Linux Foundation / CNCF

EDUCATION

Bachelor's Degree, Computer Engineering

State Technical University (2011 - 2015)

TECHNICAL SKILLS

Programming Languages: Go, Java, Python, JavaScript, TypeScript, SQL, PHP

Frameworks & Runtimes: Spring Boot, Node.js, React, Express.js

Data & Messaging: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Apache Kafka

Cloud & Infrastructure: Amazon Web Services (AWS), Google Cloud Platform (GCP), Kubernetes, Docker, Terraform

DevOps & CI/CD: CI/CD Pipelines, GitHub Actions, Jenkins, Source Control Management

Architecture & Design: Distributed Systems, Microservices, Event-Driven Architecture, REST APIs, API Design, System Design

Observability & Reliability: OpenTelemetry, Prometheus, Grafana, Incident Response

Testing: Test-Driven Development (TDD), Contract Testing, JUnit, Jest, Pytest

Leadership & Practices: Technical Mentoring, Architecture Review, Agile / Scrum, Cross-Team Collaboration

Collaboration & Productivity Tools: Slack, Jira, Confluence